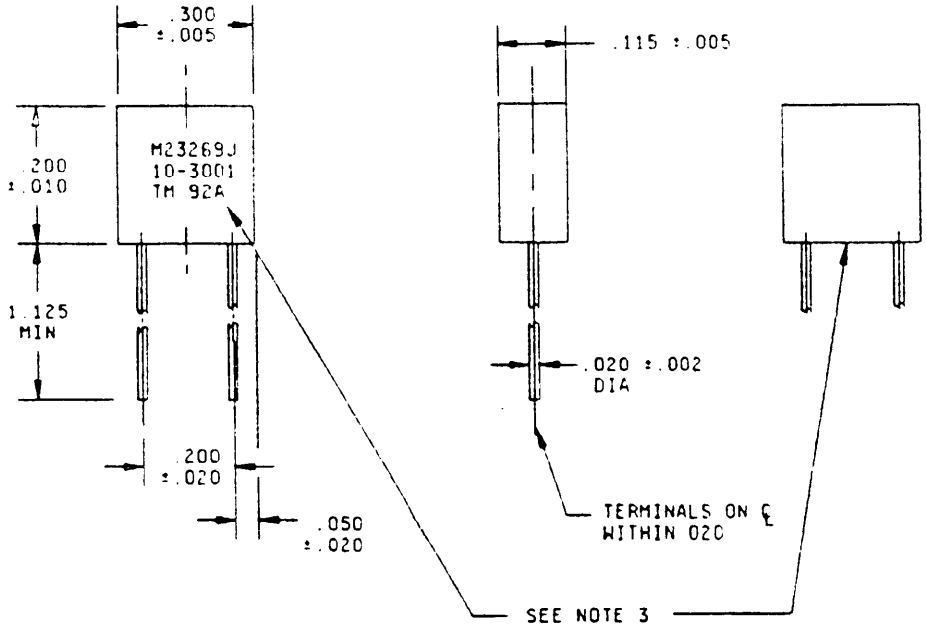
INCH-POUND
MIL-C-23269/10B
1 February 1993
SUPERSEDING
MIL-C-23269/10A
30 September 1970

MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, GLASS DIELECTRIC, (RADIAL OR AXIAL WIRE-LEAD TERMINALS), ESTABLISHED RELIABILITY, STYLES CYR51, CYR52, AND CYR53

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-23269.



STYLE CYR51, SEE TABLE I

Inches	mm
.002	0.05
.005	0.13
.010	0.25
.020	0.51
.050	1.27
.115	2.92
.200	5.08
.300	7.62
1.25	31.75

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. See marking requirement.

FIGURE 1. Dimensions and configuration.

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figures 1, 2, and 3 and tables I, II, and III.

Case material: Type D2 (gold plated copper clad iron-nickel alloy) of MIL-STD-1276. Type D5 (hot solder dip copper clad iron-nickel alloy) of MIL-STD-1276 may be furnished when specified in the contract or purchase order.

Capacitance (cap.) (nom): See tables I, II, and III.

Cap. tolerance: See tables I, II, and III.

Rated voltage: 300 volts, dc.

Failure rate level: M, P, R, or S.

Rated temperature: -55°C to +125°C.

Temperature coefficient: 140 ±25 parts/million/°C.

Capacitance drift: 0.1 percent or 0.1 pF, whichever is greater.

Seal: Test II.

Insulation resistance: 10,000 megohms (qualification inspection only at +125°C). 100,000 megohms minimum (at +25°C).

Dissipation factor (DF): 0.2 percent maximum (1.0 through 100 pF) or 0.1 percent maximum (110 pF and up).

Thermal shock and moisture resistance, qualification and group C inspection:

DF: 0.3 percent maximum (1 through 100 pF) or 0.1 percent maximum (110 pF and up).

Thermal shock, group A inspection: Capacitance shall be within the tolerance specified (see table I). DF shall be within the initial limits.

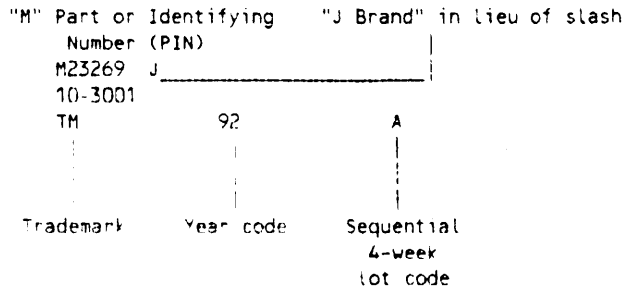
Quality factor (Q): 500 minimum (1.0 through 100 pF) or 1,000 minimum (110 pF and up).

Life:

ΔCap. (240 hours and 2,000 hours): 0.5 percent or 0.5 pF, maximum.

DF (240 hours and 2,000 hours): 0.45 percent maximum (1.0 through 100 pF) or 0.25 percent maximum (110 pF and up).

Marking:



Substitutability data: See table IV.

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TABLE I. Electrical and physical characteristics and dash numbers

Capacitance (pF)	Rated voltage (volts, dc)	Capacitance tolerance	Dash number M23269/10- FR level in percent/1,000 hours			
			M (1.0)	P (0.1)	R (0.01)	S (0.001)
1	300	±0.25 pF	3001	4001	5001	6001
1.5	"	±0.25 pF	3002	4002	5002	6002
2.2	"	±0.25 pF	3003	4003	5003	6003
2.7	"	±0.25 pF	3004	4004	5004	6004
3.0	"	±0.25 pF	3005	4005	5005	6005
3.3	"	±0.25 pF	3006	4006	5006	6006
3.6	"	±0.25 pF	3007	4007	5007	6007
3.9	"	±0.25 pF	3008	4008	5008	6008
4.3	"	±0.25 pF	3009	4009	5009	6009
4.7	"	±0.25 pF	3010	4010	5010	6010
5.1	"	±0.25 pF	3011	4011	5011	6011
5.1	"	±5%	3012	4012	5012	6012
5.6	"	±0.25 pF	3013	4013	5013	6013
5.6	"	±5%	3014	4014	5014	6014
6.2	"	±0.25 pF	3015	4015	5015	6015
6.2	"	±5%	3016	4016	5016	6016
6.8	"	±0.25 pF	3017	4017	5017	6017
6.8	"	±5%	3018	4018	5018	6018
7.5	"	±0.25 pF	3019	4019	5019	6019
7.5	"	±5%	3020	4020	5020	6020
8.2	"	±0.25 pF	3021	4021	5021	6021
8.2	"	±5%	3022	4022	5022	6022
9.1	"	±0.25 pF	3023	4023	5023	6023
9.1	"	±5%	3024	4024	5024	6024
10	"	±0.25 pF	3025	4025	5025	6025
10	"	±5%	3026	4026	5026	6026
11	"	±0.25 pF	3027	4027	5027	6027
11	"	±5%	3028	4028	5028	6028
12	"	±0.25 pF	3029	4029	5029	6029
12	"	±5%	3030	4030	5030	6030
13	"	±0.25 pF	3031	4031	5031	6031
13	"	±2%	3032	4032	5032	6032
13	"	±5%	3033	4033	5033	6033
15	"	±0.25 pF	3034	4034	5034	6034
15	"	±2%	3035	4035	5035	6035
15	"	±5%	3036	4036	5036	6036
16	"	±0.25 pF	3037	4037	5037	6037
16	"	±2%	3038	4038	5038	6038
16	"	±5%	3039	4039	5039	6039
18	"	±0.25 pF	3040	4040	5040	6040
18	"	±2%	3041	4041	5041	6041
18	"	±5%	3042	4042	5042	6042
20	"	±0.25 pF	3043	4043	5043	6043
20	"	±2%	3044	4044	5044	6044
20	"	±5%	3045	4045	5045	6045
22	"	±0.25 pF	3046	4046	5046	6046
22	"	±2%	3047	4047	5047	6047
22	"	±5%	3048	4048	5048	6048
24	"	±0.25 pF	3049	4049	5049	6049
24	"	±2%	3050	4050	5050	6050
24	"	±5%	3051	4051	5051	6051

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TABLE I. Electrical and physical characteristics and dash numbers - Continued.

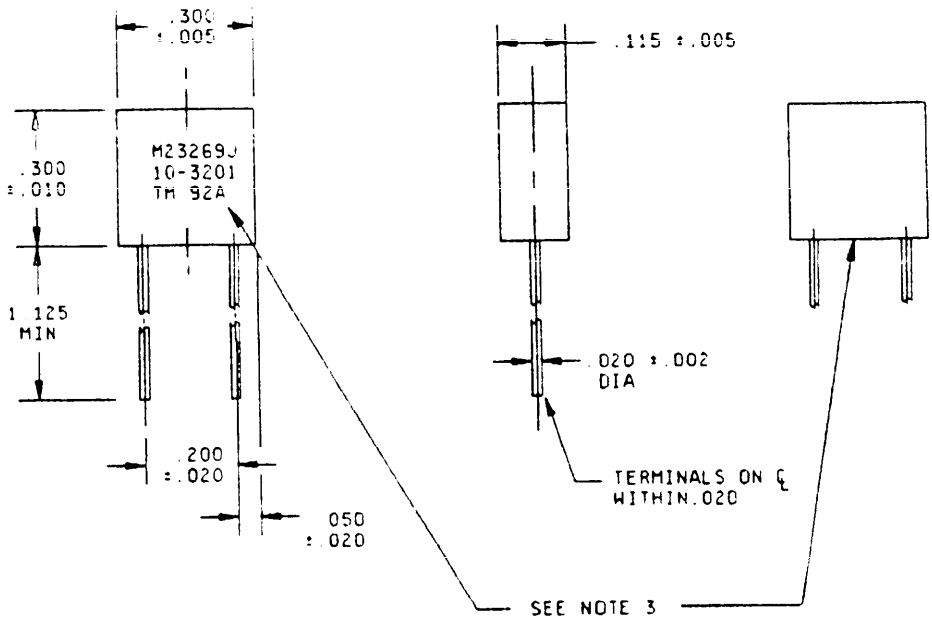
Capacitance (pF)	Rated voltage (volts, dc)	Capacitance tolerance	Dash number M23269/10- FR level in percent/1,000 hours			
			M (1.0)	P (0.1)	R (0.01)	S (0.001)
27	300	±1%	3052	4052	5052	6052
27	"	±2%	3053	4053	5053	6053
27	"	±5%	3054	4054	5054	6054
30	"	±1%	3055	4055	5055	6055
30	"	±2%	3056	4056	5056	6056
30	"	±5%	3057	4057	5057	6057
33	"	±1%	3058	4058	5058	6058
33	"	±2%	3059	4059	5059	6059
33	"	±5%	3060	4060	5060	6060
36	"	±1%	3061	4061	5061	6061
36	"	±2%	3062	4062	5062	6062
36	"	±5%	3063	4063	5063	6063
39	"	±1%	3064	4064	5064	6064
39	"	±2%	3065	4065	5065	6065
39	"	±5%	3066	4066	5066	6066
43	"	±1%	3067	4067	5067	6067
43	"	±2%	3068	4068	5068	6068
43	"	±5%	3069	4069	5069	6069
47	"	±1%	3070	4070	5070	6070
47	"	±2%	3071	4071	5071	6071
47	"	±5%	3072	4072	5072	6072
51	"	±1%	3073	4073	5073	6073
51	"	±2%	3074	4074	5074	6074
51	"	±5%	3075	4075	5075	6075
56	"	±1%	3076	4076	5076	6076
56	"	±2%	3077	4077	5077	6077
56	"	±5%	3078	4078	5078	6078
62	"	±1%	3079	4079	5079	6079
62	"	±2%	3080	4080	5080	6080
62	"	±5%	3081	4081	5081	6081
68	"	±1%	3082	4082	5082	6082
68	"	±2%	3083	4083	5083	6083
68	"	±5%	3084	4084	5084	6084
75	"	±1%	3085	4085	5085	6085
75	"	±2%	3086	4086	5086	6086
75	"	±5%	3087	4087	5087	6087
82	"	±1%	3088	4088	5088	6088
82	"	±2%	3089	4089	5089	6089
82	"	±5%	3090	4090	5090	6090
91	"	±1%	3091	4091	5091	6091
91	"	±2%	3092	4092	5092	6092
91	"	±5%	3093	4093	5093	6093
100	"	±1%	3094	4094	5094	6094
100	"	±2%	3095	4095	5095	6095
100	"	±5%	3096	4096	5096	6096
110	"	±1%	3097	4097	5097	6097
110	"	±2%	3098	4098	5098	6098
110	"	±5%	3099	4099	5099	6099
120	"	±1%	3100	4100	5100	6100
120	"	±2%	3101	4101	5101	6101
120	"	±5%	3102	4102	5102	6102

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TABLE I. Electrical and physical characteristics and dash numbers - continued

Capacitance (pF)	Rated voltage (volts dc)	Capacitance tolerance	Dash number M23269/10- FP Level in percent/1,000 hours			
			M (1.0)	P (0.1)	R (0.01)	S (0.001)
130	300	±1%	3103	4103	5103	6103
130	"	±2%	3104	4104	5104	6104
130	"	±5%	3105	4105	5105	6105
150	"	±1%	3106	4106	5106	6106
150	"	±2%	3107	4107	5107	6107
150	"	±5%	3108	4108	5108	6108
160	"	±1%	3109	4109	5109	6109
160	"	±2%	3110	4110	5110	6110
160	"	±5%	3111	4111	5111	6111
180	"	±1%	3112	4112	5112	6112
180	"	±2%	3113	4113	5113	6113
180	"	±5%	3114	4114	5114	6114
200	"	±1%	3115	4115	5115	6115
200	"	±2%	3116	4116	5116	6116
200	"	±5%	3117	4117	5117	6117
220	"	±1%	3118	4118	5118	6118
220	"	±2%	3119	4119	5119	6119
220	"	±5%	3120	4120	5120	6120
240	"	±1%	3121	4121	5121	6121
240	"	±2%	3122	4122	5122	6122
240	"	±5%	3123	4123	5123	6123
270	"	±1%	3124	4124	5124	6124
270	"	±2%	3125	4125	5125	6125
270	"	±5%	3126	4126	5126	6126
300	"	±1%	3127	4127	5127	6127
300	"	±2%	3128	4128	5128	6128
300	"	±5%	3129	4129	5129	6129
330	"	±1%	3130	4130	5130	6130
330	"	±2%	3131	4131	5131	6131
330	"	±5%	3132	4132	5132	6132
360	"	±1%	3133	4133	5133	6133
360	"	±2%	3134	4134	5134	6134
360	"	±5%	3135	4135	5135	6135
390	"	±1%	3136	4136	5136	6136
390	"	±2%	3137	4137	5137	6137
390	"	±5%	3138	4138	5138	6138
430	"	±1%	3139	4139	5139	6139
430	"	±2%	3140	4140	5140	6140
430	"	±5%	3141	4141	5141	6141
470	"	±1%	3142	4142	5142	6142
470	"	±2%	3143	4143	5143	6143
470	"	±5%	3144	4144	5144	6144
510	"	±1%	3145	4145	5145	6145
510	"	±2%	3146	4146	5146	6146
510	"	±5%	3147	4147	5147	6147
560	"	±1%	3148	4148	5148	6148
560	"	±2%	3149	4149	5149	6149
560	"	±5%	3150	4150	5150	6150

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STYLE CYR52,SEE TABLE II

Inches	mm	Inches	mm
.002	0.05	.115	2.92
.005	0.13	.200	5.08
.010	0.25	.300	7.62
.020	0.50	1.25	31.75
.050	1.27		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. See marking requirement.

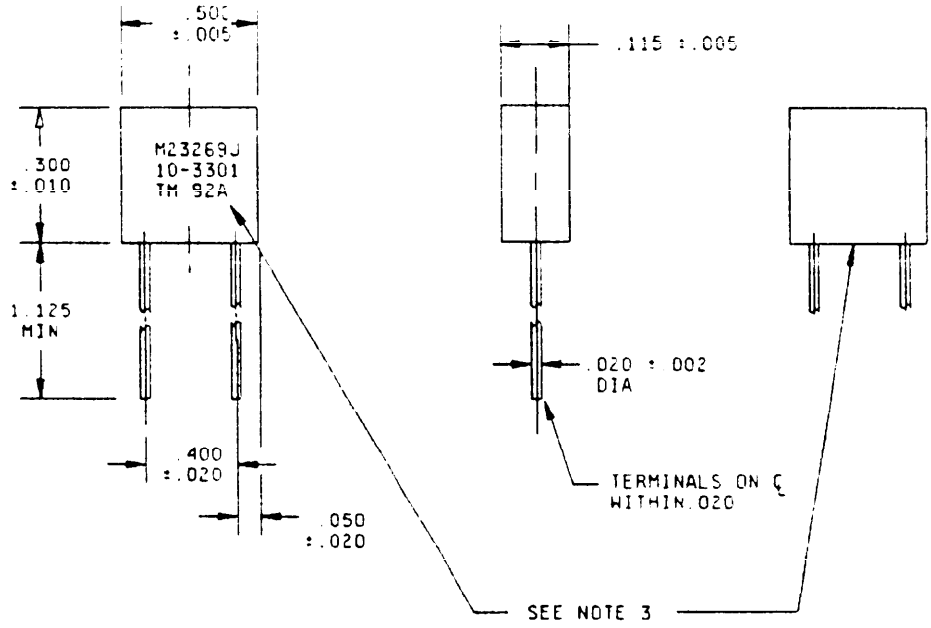
FIGURE 1. Dimensions and configuration for style CYR52.

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TABLE II. Electrical and physical characteristics and dash numbers for style CYR52.

Capacitance (pf)	Rated voltage (volts, dc)	Capacitance tolerance	Dash number M23269/10- FR level in percent/1,000 hours			
			M (1.0)	P (0.1)	R (0.01)	S (0.001)
620	300	±1%	3201	4201	5201	6101
620	"	±2%	3202	4202	5202	6102
620	"	±5%	3203	4203	5203	6103
680	"	±1%	3204	4204	5204	6104
680	"	±2%	3205	4205	5205	6105
680	"	±5%	3206	4206	5206	6106
750	"	±1%	3207	4207	5207	6107
750	"	±2%	3208	4208	5208	6108
750	"	±5%	3209	4209	5209	6109
820	"	±1%	3210	4210	5210	6110
820	"	±2%	3211	4211	5211	6111
820	"	±5%	3212	4212	5212	6112
910	"	±1%	3213	4213	5213	6113
910	"	±2%	3214	4214	5214	6114
910	"	±5%	3215	4215	5215	6115
1,000	"	±1%	3216	4216	5216	6116
1,000	"	±2%	3217	4217	5217	6117
1,000	"	±5%	3218	4218	5218	6118

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STYLE CYR53,SEE TABLE III

Inches	mm	Inches	mm
.002	0.05	.115	2.92
.005	0.13	.300	7.62
.010	0.25	.400	10.16
.020	0.50	.500	12.75
.050	1.27	1.25	31.75

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. See marking requirement.

FIGURE 2. Dimensions and configuration for style CYR53.

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TABLE III. Electrical and physical characteristics and dash numbers for style CYR53.

Capacitance (pf)	Rated voltage (volts, dc)	Capacitance tolerance	Dash number M23269/10- FR level in percent/1,000 hours			
			M (1.0)	P (0.1)	R (0.01)	S(0.001)
1,100	300	±1%	3301	4301	5301	6301
1,100	"	±2%	3302	4302	5302	6302
1,100	"	±5%	3303	4303	5303	6303
1,200	"	±1%	3304	4304	5304	6304
1,200	"	±2%	3305	4305	5305	6305
1,200	"	±5%	3306	4306	5306	6306
1,300	"	±1%	3307	4307	5307	6307
1,300	"	±2%	3308	4308	5308	6308
1,300	"	±5%	3309	4309	5309	6309
1,500	"	±1%	3310	4310	5310	6310
1,500	"	±2%	3311	4311	5311	6311
1,500	"	±5%	3312	4312	5312	6312
1,600	"	±1%	3313	4313	5313	6313
1,600	"	±2%	3314	4314	5314	6314
1,600	"	±5%	3315	4315	5315	6315
1,800	"	±1%	3316	4316	5316	6316
1,800	"	±2%	3317	4317	5317	6317
1,800	"	±5%	3318	4318	5318	6318
2,000	"	±1%	3319	4319	5319	6319
2,000	"	±2%	3320	4320	5320	6320
2,000	"	±5%	3321	4321	5321	6321
2,200	"	±1%	3322	4322	5322	6322
2,200	"	±2%	3323	4323	5323	6323
2,200	"	±5%	3324	4324	5324	6324
2,400	"	±1%	3325	4325	5325	6325
2,400	"	±2%	3326	4326	5326	6326
2,400	"	±5%	3327	4327	5327	6327

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TABLE II. Substitutability data

Superseded type designation in MIL-C-23269/10	Use dash number M23269/10- in MIL-C-23269/10A and -10B FR level, in %/1,000 hours		Superseded type designation in MIL-C-23269/10	Use dash number M23269/10- in MIL-C-23269/10A and -10B FR level, in %/1,000 hours	
	L (2.0)	M (1.0)		L (2.0)	M (1.0)
CYR51C1R0C	2001	3001	CYR51C270F	2052	3052
CYR51C1R5C	2002	3002	CYR51C270G	2053	3053
CYR51C2R2C	2003	3003	CYR51C270J	2054	3054
CYR51C2R7C	2004	3004	CYR51C300F	2055	3055
CYR51C3R0C	2005	3005	CYR51C300G	2056	3056
CYR51C3R3C	2006	3006	CYR51C300J	2057	3057
CYR51C3R6C	2007	3007	CYR51C330F	2058	3058
CYR51C3R9C	2008	3008	CYR51C330G	2059	3059
CYR51C4R3C	2009	3009	CYR51C330J	2060	3060
CYR51C4R7C	2010	3010	CYR51C360F	2061	3061
CYR51C5R1C	2011	3011	CYR51C360G	2062	3062
CYR51C5R1J	2012	3012	CYR51C360J	2063	3063
CYR51C5R6C	2013	3013	CYR51C390F	2064	3064
CYR51C5R6J	2014	3014	CYR51C390G	2065	3065
CYR51C6R2C	2015	3015	CYR51C390J	2066	3066
CYR51C6R2J	2016	3016	CYR51C430F	2067	3067
CYR51C6R8C	2017	3017	CYR51C430G	2068	3068
CYR51C6R8J	2018	3018	CYR51C430J	2069	3069
CYR51C7R5C	2019	3019	CYR51C470F	2070	3070
CYR51C7R5J	2020	3020	CYR51C470G	2071	3071
CYR51C8R2C	2021	3021	CYR51C470J	2072	3072
CYR51C8R2J	2022	3022	CYR51C510F	2073	3073
CYR51C9R1C	2023	3023	CYR51C510G	2074	3074
CYR51C9R1J	2024	3024	CYR51C510J	2075	3075
CYR51C100C	2025	3025	CYR51C560F	2076	3076
CYR51C100J	2026	3026	CYR51C560G	2077	3077
CYR51C110C	2027	3027	CYR51C560J	2078	3078
CYR51C110J	2028	3028	CYR51C620F	2079	3079
CYR51C120C	2029	3029	CYR51C620G	2080	3080
CYR51C120J	2030	3030	CYR51C620J	2081	3081
CYR51C130C	2031	3031	CYR51C680F	2082	3082
CYR51C130G	2032	3032	CYR51C680G	2083	3083
CYR51C130J	2033	3033	CYR51C680J	2084	3084
CYR51C150C	2034	3034	CYR51C750F	2085	3085
CYR52C150G	2035	3035	CYR51C750G	2086	3086
CYR51C150J	2036	3036	CYR51C750J	2087	3087
CYR51C160C	2037	3037	CYR51C820F	2088	3088
CYR51C160G	2038	3038	CYR51C820G	2089	3089
CYR51C160J	2039	3039	CYR51C820J	2090	3090
CYR51C180C	2040	3040	CYR51C910F	2091	3091
CYR51C180G	2041	3041	CYR51C910G	2092	3092
CYR51C180J	2042	3042	CYR51C910J	2093	3093
CYR51C200C	2043	3043	CYR51C101F	2094	3094
CYR51C200G	2044	3044	CYR51C101G	2095	3095
CYR51C200J	2045	3045	CYR51C101J	2096	3096
CYR51C220C	2046	3046	CYR51C111F	2097	3097
CYR51C220G	2047	3047	CYR51C111G	2098	3098
CYR51C220J	2048	3048	CYR51C111J	2099	3099
CYR51C240C	2049	3049	CYR51C121F	2100	3100

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TABLE IV. Substitutability data - Continued.

Superseded type designation in MIL-C-23269/10	Use dash number M23269/10- in MIL-C-23269/10A and /10B	
	FF level, in 1/1,000 hours	
	L (2.0)	M (1.0)
CYR51C240G	2050	3050
CYR51C240J	2051	3051
CYR51C131F	2103	3103
CYR51C131G	2104	3104
CYR51C131J	2105	3105
CYR51C151F	2106	3106
CYR51C151G	2107	3107
CYR51C151J	2108	3108
CYR51C161F	2109	3109
CYR51C161G	2110	3110
CYR51C161J	2111	3111
CYR51C181F	2112	3112
CYR51C181G	2113	3113
CYR51C181J	2114	3114
CYR51C201F	2115	3115
CYR51C201G	2116	3116
CYR51C201J	2117	3117
CYR51C221F	2118	3118
CYR51C221G	2119	3119
CYR51C221J	2120	3120
CYR51C241F	2121	3121
CYR51C241G	2122	3122
CYR51C241J	2123	3123
CYR51C271F	2124	3124
CYR51C271G	2125	3125
CYR51C271J	2126	3126
CYR51C301F	2127	3127
CYR51C301G	2128	3128
CYR51C301J	2129	3129
CYR51C331F	2130	3130
CYR51C331G	2131	3131
CYR51C331J	2132	3132
CYR51C361F	2133	3133
CYR51C361G	2134	3134
CYR51C361J	2135	3135
CYR51C391F	2136	3136
CYR51C391G	2137	3137
CYR51C391J	2138	3138
CYR51C431F	2139	3139
CYR51C431G	2140	3140
CYR51C431J	2141	3141
CYR51C471F	2142	3142
CYR51C471G	2143	3143
CYR51C471J	2144	3144
CYR51C511F	2145	3145
CYR51C511G	2146	3146
CYR51C511J	2147	3147
CYR51C561F	2148	3148
CYR51C561G	2149	3149
CYR51C561J	2150	3150

Superseded type designation in MIL-C-23269/10	Use dash number M23269/10- in MIL-C-23269/10A and /10B	
	FF level, in 1/1,000 hours	
	L (2.0)	M (1.0)
CYR51C121G	2101	3101
CYR51C121J	2102	3102
CYR51C621F	2201	3201
CYR51C621G	2202	3202
CYR51C621J	2203	3203
CYR51C681F	2204	3204
CYR51C681G	2205	3205
CYR51C681J	2206	3206
CYR51C751F	2207	3207
CYR51C751G	2208	3208
CYR51C751J	2209	3209
CYR51C821F	2210	3210
CYR51C821G	2211	3211
CYR51C821J	2212	3212
CYR51C911F	2213	3213
CYR51C911G	2214	3214
CYR51C911J	2215	3215
CYR51C102F	2216	3216
CYR51C102G	2217	3217
CYR51C102J	2218	3218
CYR51C112F	2301	3301
CYR51C112G	2302	3302
CYR51C112J	2303	3303
CYR51C122F	2304	3304
CYR51C122G	2305	3305
CYR51C122J	2306	3306
CYR51C132F	2307	3307
CYR51C132G	2308	3308
CYR51C132J	2309	3309
CYR51C152F	2310	3310
CYR51C152G	2311	3311
CYR51C152J	2312	3312
CYR51C162F	2313	3313
CYR51C162G	2314	3314
CYR51C162J	2315	3315
CYR51C182F	2316	3316
CYR51C182G	2317	3317
CYR51C182J	2318	3318
CYR51C202F	2319	3319
CYR51C202G	2320	3320
CYR51C202J	2321	3321
CYR51C222F	2322	3322
CYR51C222G	2323	3323
CYR51C222J	2324	3324
CYR51C242F	2325	3325
CYR51C242G	2326	3326
CYR51C242J	2327	3327

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Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Air Force - 17
DLA - ES

User activities:
Navy - AS, CG, MC, OS, SH
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

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